SINGAPORE DIMENSIONS • GRADE 2A • CHAPTER 4

# Multiplication & Doubles: 2, 5 & 10 Facts

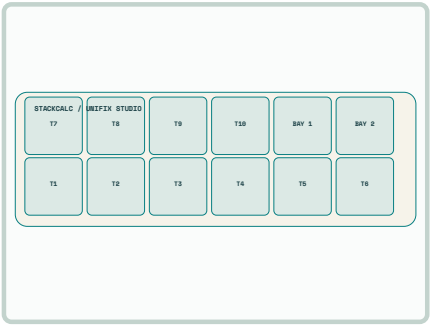
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.2.OA.C.4

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

**STUDENT SELF-CHECK RULE (NO CHEATING):**
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

## STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

### 1. 2D Bench Blueprint




Kit: Unifix Tray

### 2. Desk Setup (Concrete -> Pictorial)

**Physical Bench Setup:**
Place double dominoes (4|4, 6|6). Build 2-column cube arrays.

**Sketch Model on Paper:**
Draw an array grid: 2 rows of 6 dots.

### 3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1. 8 **Step 1** Execute [8]
2. enter **Step 2** Execute [enter]
3. + **Step 3** Execute [plus]

EXPECTED LCD X: 16

**1** **Double 7: 7 ENTER +**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**2** **Double 9: 9 ENTER +**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**3** **3 groups of 5: [5] [ENTER] [3] [×]**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**4** **4 groups of 10: [10] [ENTER] [4] [×]**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

# Station Diagnostic & Solution Key: Chapter 4

Multiplication & Doubles: 2, 5 & 10 Facts • Manipulative Setup, Diagnostic Rubric & Verification Flow

## MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

**Station Kit: Math Dominoes & Interlocking Cubes**

CAD Blueprint: unifix_tray

**Target Device: SC-6 Elementary**

**CAD / STL Source File:**

Education/scad/unifix_cubes_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

**Recommended Filaments:**

MultiBuild Interlocking Cubes in Rainbow PLA

| # | Math Problem Prompt | Expected Math Answer | Self-Check Keystrokes & RPN Stack |
|---|---|---|---|
| 1 | **Double 7: 7 ENTER +**<br>Standard: CCSS.MATH.2.OA.C.4 | 14 | [7] [ENTER] [+] |
| 2 | **Double 9: 9 ENTER +**<br>Standard: CCSS.MATH.2.OA.C.4 | 18 | [9] [ENTER] [+] |
| 3 | **3 groups of 5: [5] [ENTER] [3] [×]**<br>Standard: CCSS.MATH.2.OA.C.4 | 15 | [5] [ENTER] [3] [×] |
| 4 | **4 groups of 10: [10] [ENTER] [4] [×]**<br>Standard: CCSS.MATH.2.OA.C.4 | 40 | [1][0] [ENTER] [4] [×] |

## TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

**Key Misconception to Target During Station Observation:**

Distinguish 2 rows of 5 from 5 rows of 2.

**How the Manipulatives & Calculator Foster Self-Correction:**

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

**Station Facilitation & Pacing Tip:**

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

## CLASSROOM SCORING & STATION MASTERY RUBRIC

| | |
|---|---|
| **Exemplary (18–20 pts):** | Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack. |
| **Proficient (14–17 pts):** | Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting. |
| **Developing (10–13 pts):** | Understands problem concept but made computational slips; successfully self-corrected using calculator LCD. |
| **Intervention (<10 pts):** | Relied on calculator before solving on paper, or struggled with underlying physical manipulative models. |